

MACHA, Russian Federation

WMO: 300740

Lat: 59.904N

Lon: 117.588E

Elev: 164

StdP: 99.37

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-46.9	-44.3	-50.4	0.0	-41.4	-47.0	0.0	-44.2	9.0	-14.6	6.5	-15.1	0.2	330	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	29.3	18.6	26.8	17.7	24.6	16.7	20.1	26.8	18.9	24.7	17.8	22.8	2.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.9	13.1	22.5	16.7	12.2	21.3	15.7	11.4	20.1	58.4	26.9	54.7	24.7	51.2	23.0	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.8	6.5	5.5	-47.6	32.8	2.1	2.1	-49.0	34.4	-50.2	35.6	-51.4	36.8	-52.9	38.4		
			-47.9	21.8	3.5	1.5	-50.4	22.9	-52.4	23.8	-54.4	24.6	-56.9	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.5	-29.9	-24.1	-12.3	-1.3	6.9	15.0	17.5	15.3	7.1	-3.1	-19.0	-27.5
	DBStd	18.01	9.29	8.13	8.09	5.41	4.64	3.74	3.44	3.57	3.96	5.63	8.74	9.82
	HDD10.0	5895	1236	956	692	339	119	4	0	4	106	404	871	1164
	HDD18.3	8398	1494	1189	950	589	354	111	57	108	338	663	1121	1422
	CDD10.0	596	0	0	0	0	23	155	233	167	17	0	0	0
	CDD18.3	56	0	0	0	0	0	12	32	12	0	0	0	0
	CDH23.3	718	0	0	0	0	24	193	365	134	2	0	0	0
	CDH26.7	208	0	0	0	0	5	47	121	36	0	0	0	0
Wind	WSAvg	1.9	1.2	1.5	2.1	2.6	2.6	2.2	1.9	1.7	2.0	2.3	1.6	1.4
Precipitation	PrecAvg	479	17	14	12	22	46	64	88	77	56	34	27	20
	PrecMax	624	29	34	23	55	92	107	153	156	99	53	43	35
	PrecMin	319	7	5	3	10	26	35	35	24	25	17	8	8
	PrecStd	69	5	6	4	9	17	19	36	24	17	11	9	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.1	-3.6	8.8	13.9	26.3	31.0	33.0	30.7	21.5	11.5	6.1	-2.1
		MCWB	-9.6	-5.7	3.4	6.7	14.3	17.7	20.0	18.9	14.0	6.1	3.3	-4.1
	2%	DB	-10.5	-6.2	4.4	11.1	22.0	28.3	30.0	26.8	17.9	8.7	-0.9	-5.0
		MCWB	-11.5	-7.6	0.4	4.5	11.9	16.9	19.3	18.4	12.1	5.0	-2.6	-6.1
	5%	DB	-13.0	-8.7	1.6	9.2	19.0	25.9	27.4	24.4	15.4	6.3	-4.3	-8.2
		MCWB	-13.8	-9.7	-1.6	3.4	10.0	16.4	18.9	17.1	10.9	2.9	-5.8	-9.1
	10%	DB	-16.2	-12.3	-0.9	7.0	15.7	23.7	24.9	22.3	13.3	4.2	-6.9	-12.5
		MCWB	-16.7	-13.2	-3.2	2.2	8.4	15.2	17.7	16.2	9.6	1.6	-7.9	-13.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.1	-5.6	3.3	7.2	14.7	20.2	22.0	20.9	14.6	6.8	3.2	-3.8
		MCDB	-8.3	-3.9	8.4	12.5	26.3	27.2	28.7	26.5	20.6	10.2	6.5	-2.4
	2%	WB	-11.4	-7.8	0.6	5.1	12.3	18.6	20.7	19.5	13.1	5.2	-2.5	-6.1
		MCDB	-10.5	-6.6	4.3	10.5	21.1	25.5	28.2	24.8	16.7	8.3	-0.7	-4.8
	5%	WB	-13.9	-9.9	-1.4	3.8	10.5	17.4	19.5	18.3	11.6	3.4	-5.8	-8.9
		MCDB	-13.1	-8.8	1.3	8.3	17.7	23.8	26.0	22.9	14.5	6.0	-4.5	-8.1
	10%	WB	-16.7	-13.4	-3.1	2.5	8.9	16.1	18.4	17.2	10.1	1.8	-7.9	-12.8
		MCDB	-16.1	-12.6	-0.8	6.7	14.8	22.1	23.7	21.4	12.9	4.2	-6.9	-12.2
Mean Daily Temperature Range	5% DB	MDBR	8.2	10.4	13.0	11.1	11.0	12.1	10.9	10.9	9.1	6.7	7.8	7.4
		MCDBR	10.5	12.1	12.9	13.4	17.6	17.0	16.3	16.2	13.6	9.0	10.1	9.0
		MCWBR	9.6	11.4	9.3	7.9	8.8	7.4	7.0	7.9	7.7	6.1	8.7	8.1
	5% WB	MCDBR	10.3	12.2	12.0	11.6	16.4	13.1	13.3	12.4	11.8	8.4	9.6	9.3
		MCWBR	9.5	11.1	8.8	7.1	8.6	6.4	6.4	6.3	7.0	5.8	8.4	8.4
Clear-Sky Solar Irradiance	taub	0.250	0.284	0.311	0.355	0.393	0.434	0.481	0.427	0.340	0.297	0.253	0.224	
	taud	1.976	2.027	2.062	2.124	2.252	2.230	2.094	2.229	2.448	2.356	2.036	1.921	
	Ebn at Noon	493	693	803	835	832	799	742	758	788	702	487	368	
	Edh at Noon	49	83	111	127	121	127	142	115	78	61	45	34	
All-Sky Solar Radiation	RadAvg	0.47	1.37	2.97	4.71	5.04	5.52	5.00	3.95	2.47	1.22	0.59	0.26	
	RadStd	0.03	0.09	0.19	0.28	0.33	0.52	0.52	0.24	0.24	0.11	0.07	0.02	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page